

# GEOHAPPEN

A Geo Location based Event Finder

GUIDE: MR. AJITH S

TEAM 2:

ALEENA SHAJAN (U2201023)  
ANAGHA M ANIL (U2201031)  
ANGELINA MARY GIBY(U2201034)  
ARCHITHA B (U2201051)

# INTRODUCTION

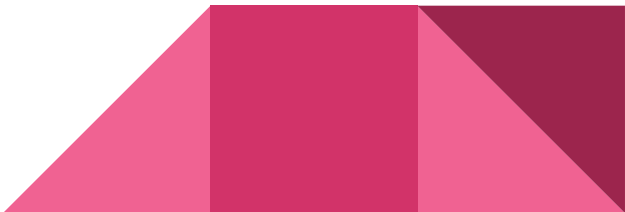
- ❑ GeoHappen is an application that offers a centralized platform where users, with access to the internet, can instantly discover events based on their current location.
  - ❑ Events include college fests, workshops, concerts and public gatherings.
  - ❑ This project helps users stay informed and engaged about different events.
  - ❑ By integrating geolocation and cloud technologies, GeoHappen enhances real-world participation and connects people to events of their interest.
- 

# PROBLEM STATEMENT

- Users miss nearby events due to lack of real-time, location-based discovery tools.
- Event organizers struggle to reach the right audience in their area.
- Manual event search is inefficient and time-consuming.



# OBJECTIVES

- To develop a location-aware system that helps users find events happening nearby in real-time.
  - Integrate at least 12 events into the database before launch.
  - Ensure event search results load within 3 seconds after the user's location is detected.
  - Allow filtering by date and category and also provide recommendations.
- 

# SYSTEM OVERVIEW

## User Interface

- Filtering by date and Category
- Display event details
- User registration

## Backend Features

- Event data management
- Geocoding and Reverse Geocoding
- Data validation



# SYSTEM OVERVIEW

## Database

- Store user profiles
- Store event data

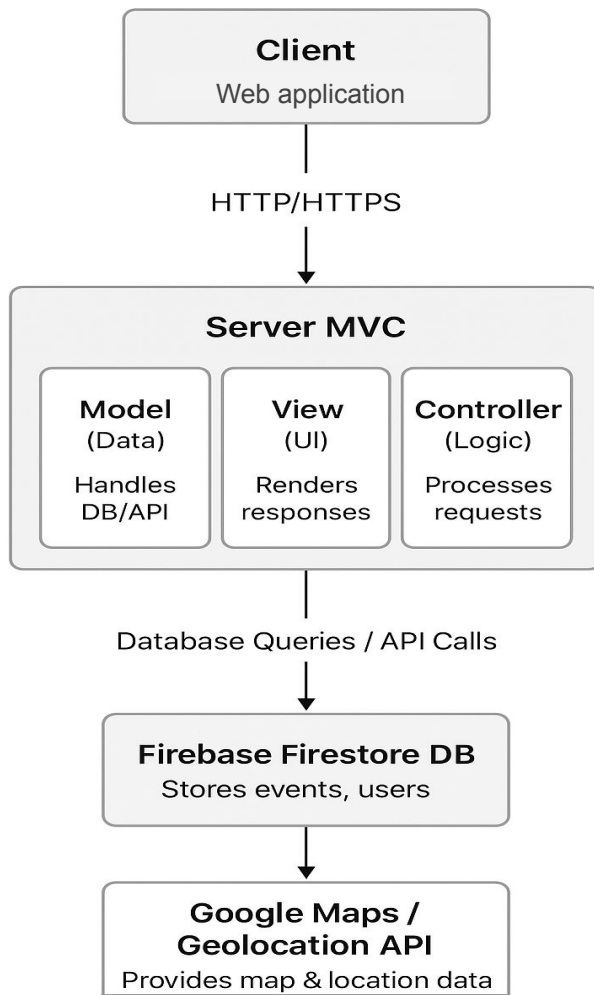
## Location Services

- GPS data from user's device

## Target Audience

- General public
- Event organizers



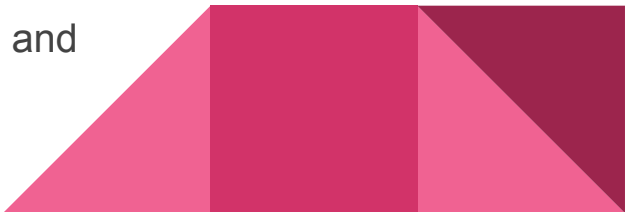


# SYSTEM ARCHITECTURE

Client–Server MVC  
(Model-View-Controller) Flow

# Explanation of Layers

- **Client (View Layer)**
  - **Node.js** renders the UI in the browser.
  - Users interact via event listings, map, filters, and recommendations.
  - Sends HTTP requests to the backend.
- **Controller (Application Logic)**
  - Part of the backend in **Node.js**.
  - Handles incoming requests, processes data, calls models, and selects the view for desired event.





# Explanation of Layers

- **Model (Data Layer)**

- Interacts with **Firebase Firestore** to store and retrieve events, user profiles, and organizer details.
- Fetches external data from **Google Maps API**.

- **View (Presentation Layer)**

- Returns HTML/JSON data to the client, which Node.js renders dynamically.



# MODULE DESIGN

## 1. UI (User Interface) Module

- Home screen containing a list of nearby events
- A map showing event location
- Event details screen
- Filters and categories
- User profile

## 2. UX (User Experience) Module

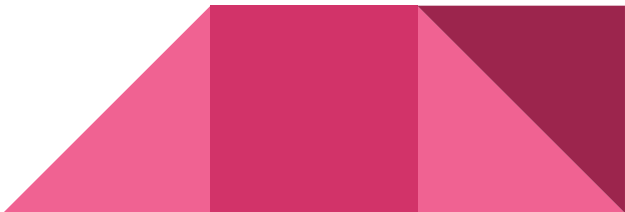
- Fallback messages displayed in case of no events found or GPS failure

# MODULE DESIGN

## 3. Models used

- Event model
  - event\_id, title, description, category, location, latitude, longitude, start\_time, end\_time
- User model
  - user\_id, name, email, preferences
- Location model
  - latitude, longitude, address

## 4. Communication interfaces used

- Geolocation services
  - Mapping and navigation APIs
  - Event data APIs
  - Backend communication
- 

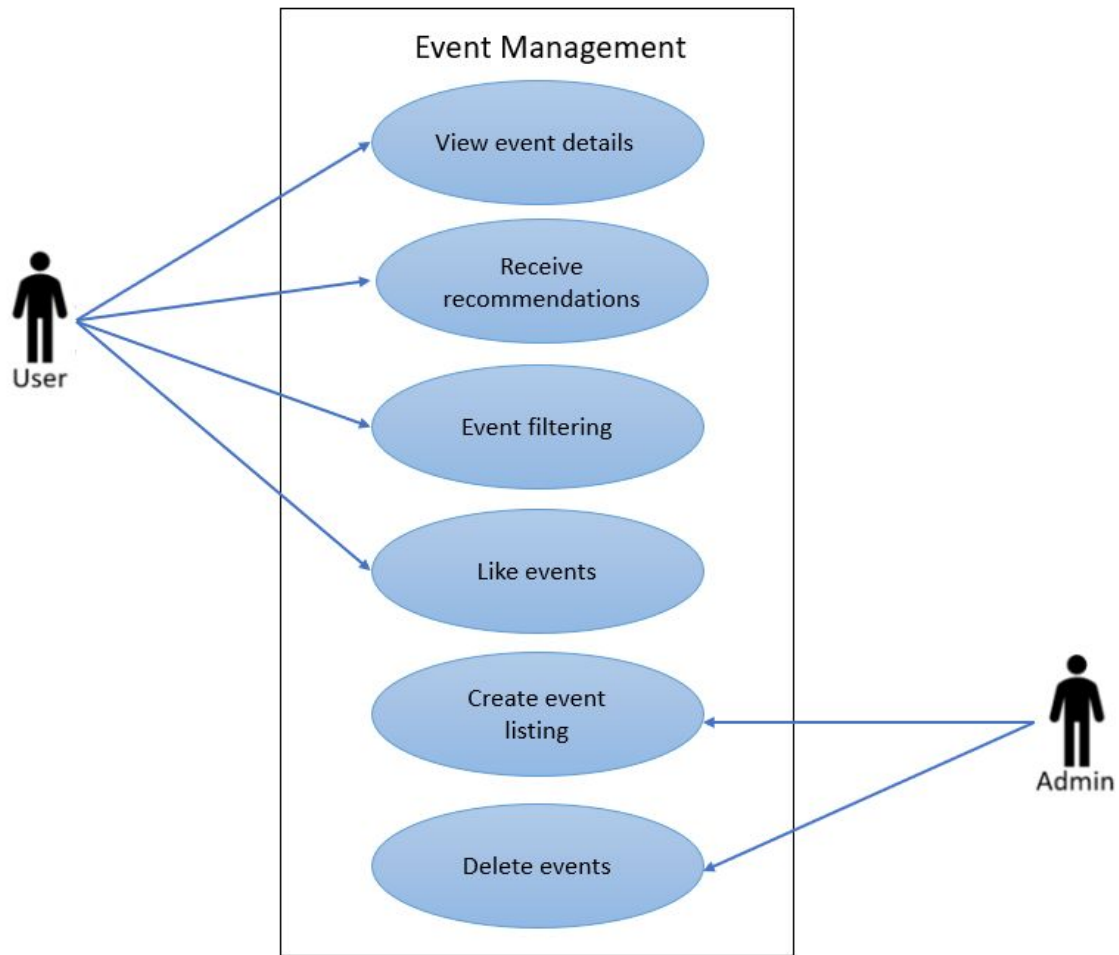
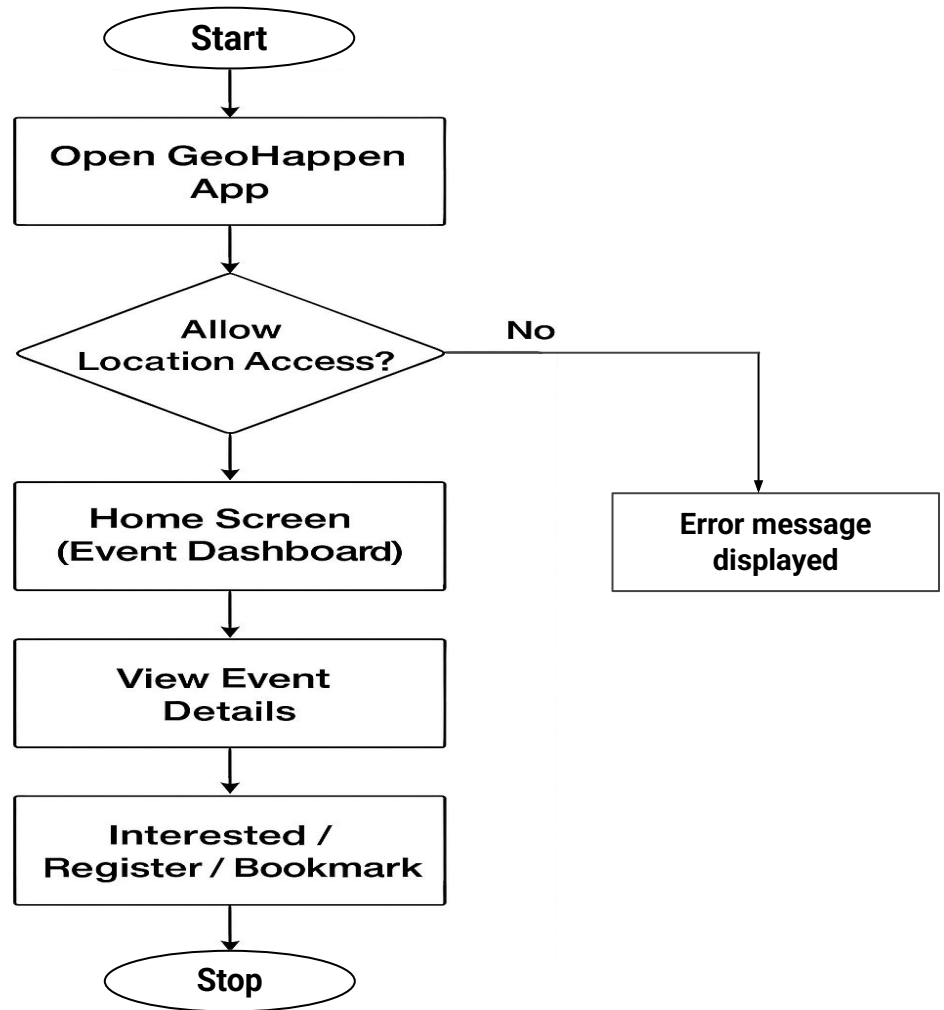
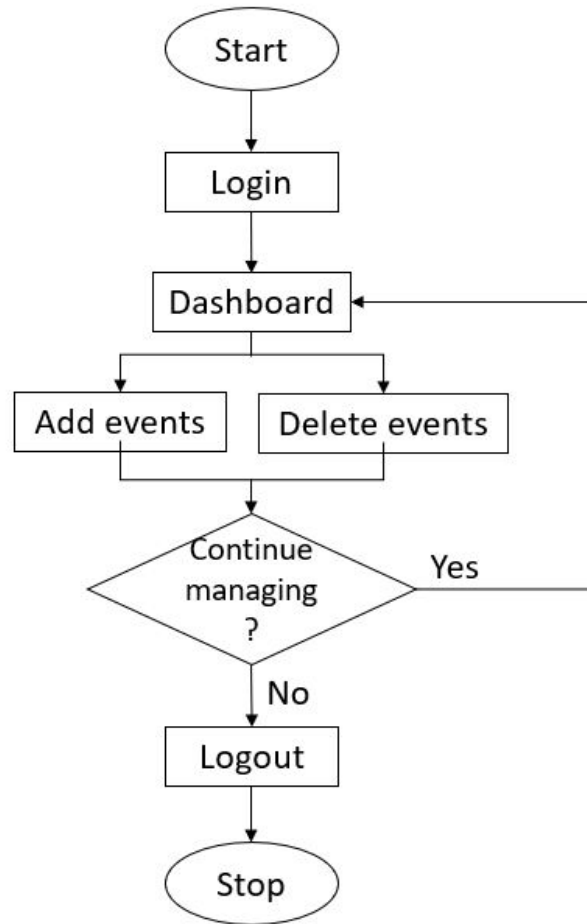


Figure 1: Use Case Diagram

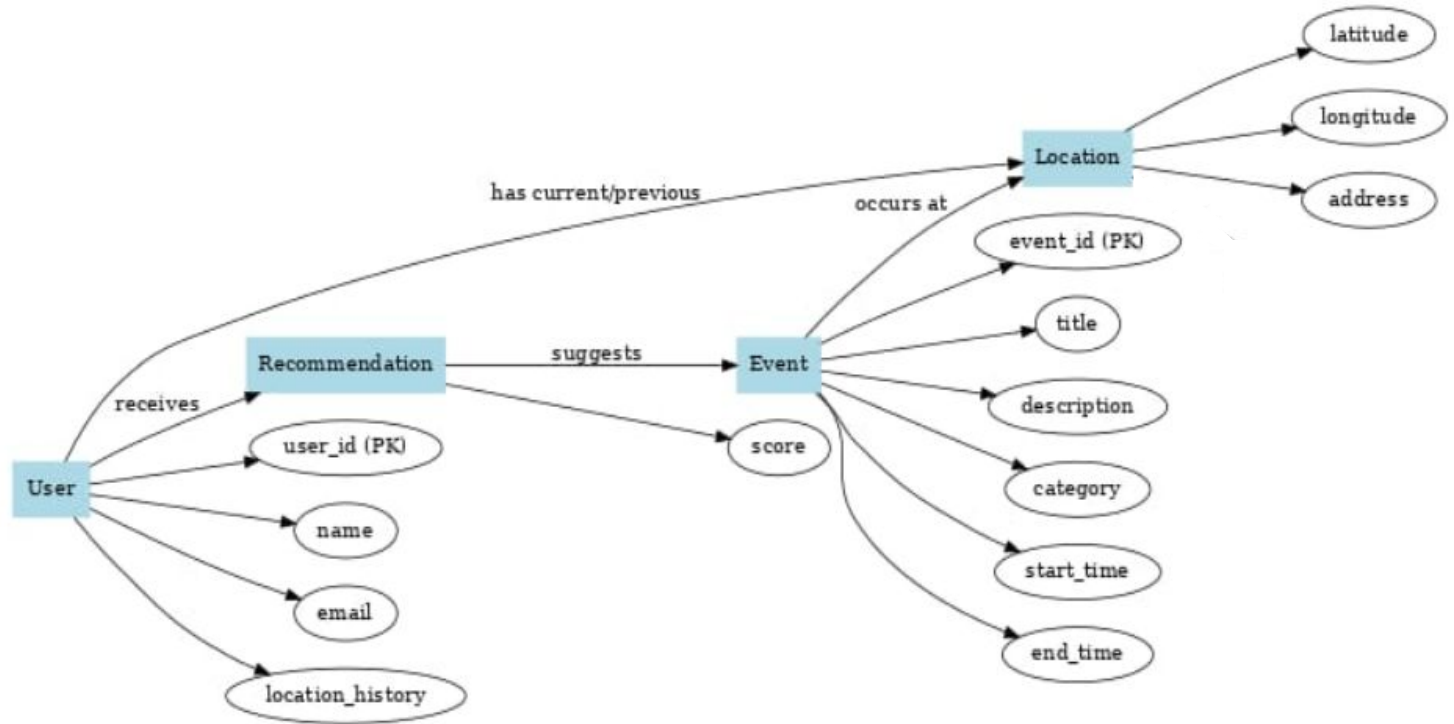
# USER SIDE FLOWCHART

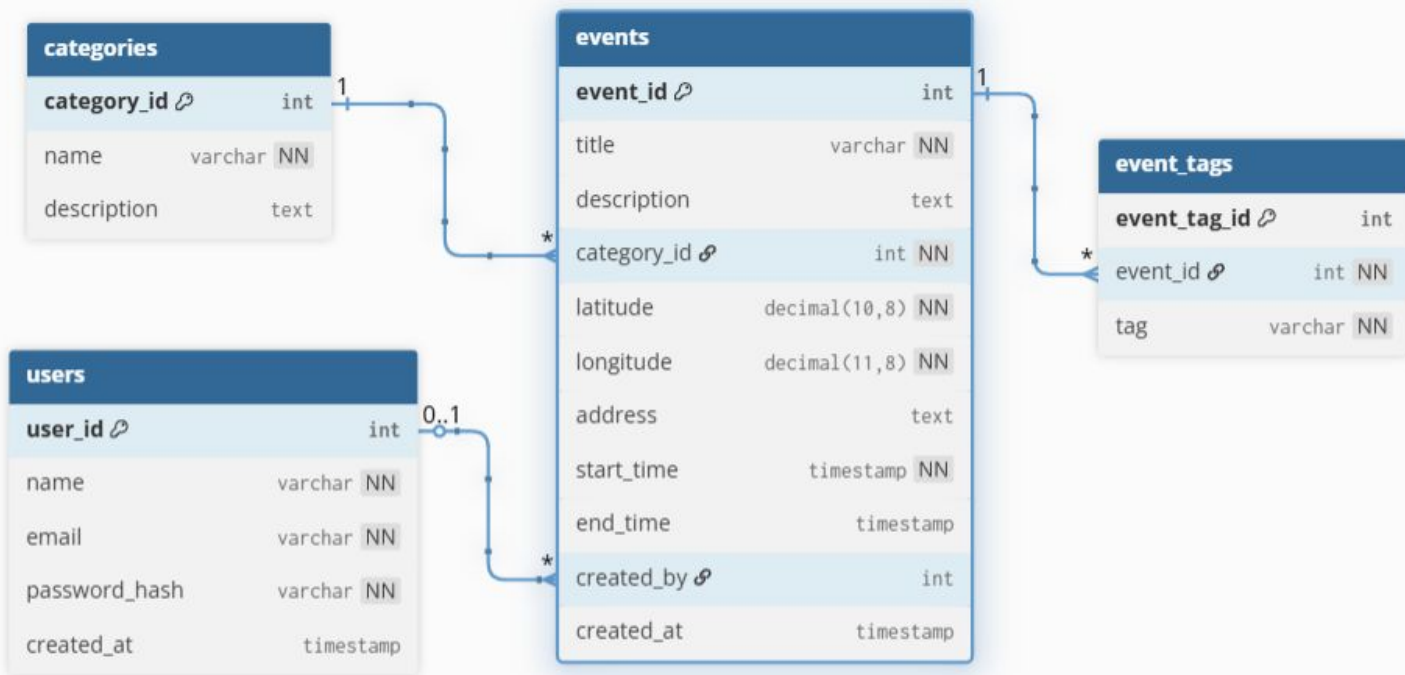




## ADMIN SIDE FLOWCHART

## ER Diagram

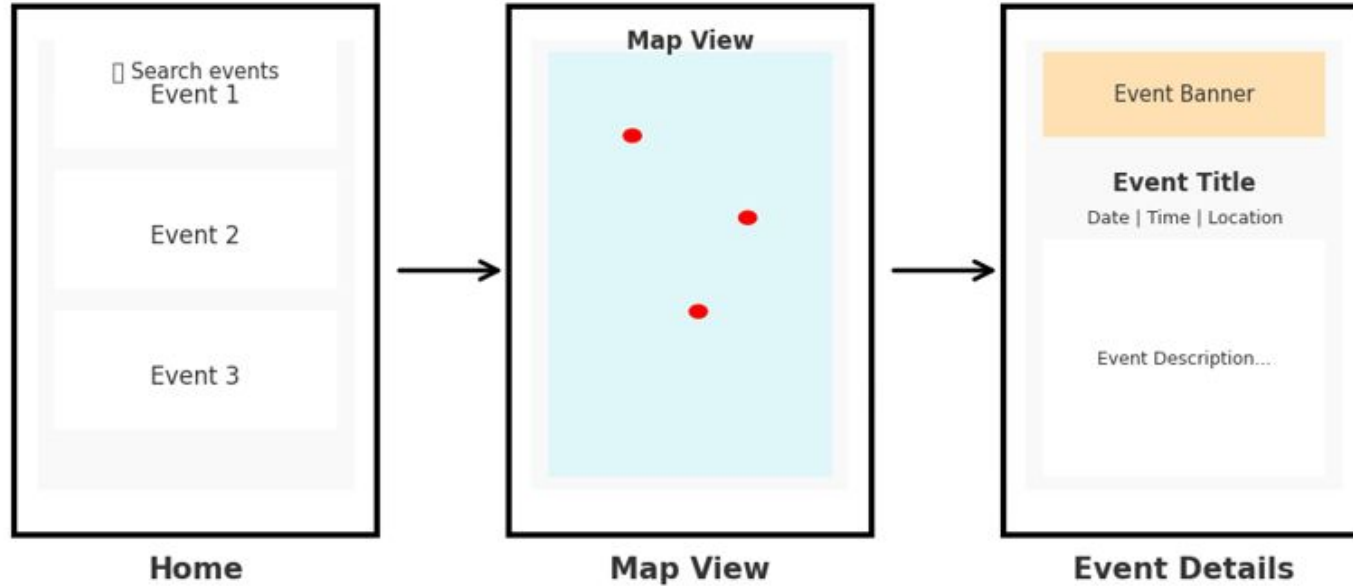




## Database Design



# USER INTERFACE DESIGN

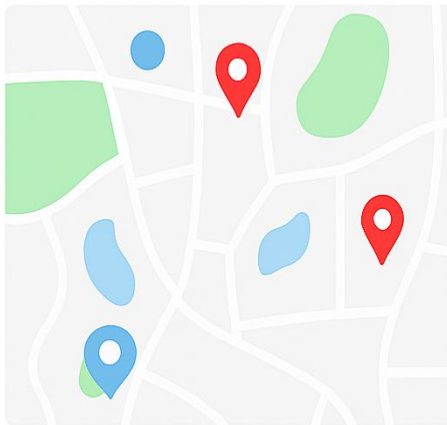


Mockup Diagram

# Sample Home Page Interface

[Home](#)[Events](#)

## Discover Events



### Community Meetup

12:30 pm - 2:00 pm

Central Park

### Art Exhibition

4:00 pm - 7:00 pm

City Gallery

### Live Music Concert

6:00 pm - 8:30 pm

Downtown Arena

- ❖ **Header Section** – Contains the app name/logo *GeoHappen* and quick navigation links (Home, Events, User profile,).
- ❖ **Filters** – Allows users to filter events based on category and time period.
- ❖ **Interactive Map (Google Maps API)** – Displays event markers based on user and event location .
- ❖ **Event Cards/Listing Section** – Shows upcoming events with title, date, time and location along with details about the host.
- ❖ **Interaction with Cards** – Users can like events and hosts can delete events created by them.
- ❖ **Receive Recommendations** – Users will receive recommendations based on the events they liked.
- ❖ **User Location Indicator** – Highlights the current position of the user on the map for easy navigation



# Login page Interface



HomeEventsAboutContactLogin / Sign Up

## Login

Don't have an account? [Sign up](#)

Email

Email

Password

Password

☐ Remember me[Forgot password?](#)

Login

# Dashboard Interface



HomeEventsAboutContactLogout

Dashboard

Events

Profile

## Welcome!

Here are some upcoming events near you

### Upcoming Events



Music Concert  
April 27, 2024  
8:00 PM



Community Meetup  
May 2, 2024  
3:00 PM

# TOOLS AND TECHNOLOGIES USED

- **Frontend (View):** HTML, CSS, JavaScript
- **Backend (Controller & Model):** Node.js
- **Database:** Firebase Firestore (real-time DB)
- **External APIs:** Google Maps JavaScript API, Geolocation API
- **Tools:** VS Code



# CHALLENGES AND MITIGATION

## Challenges

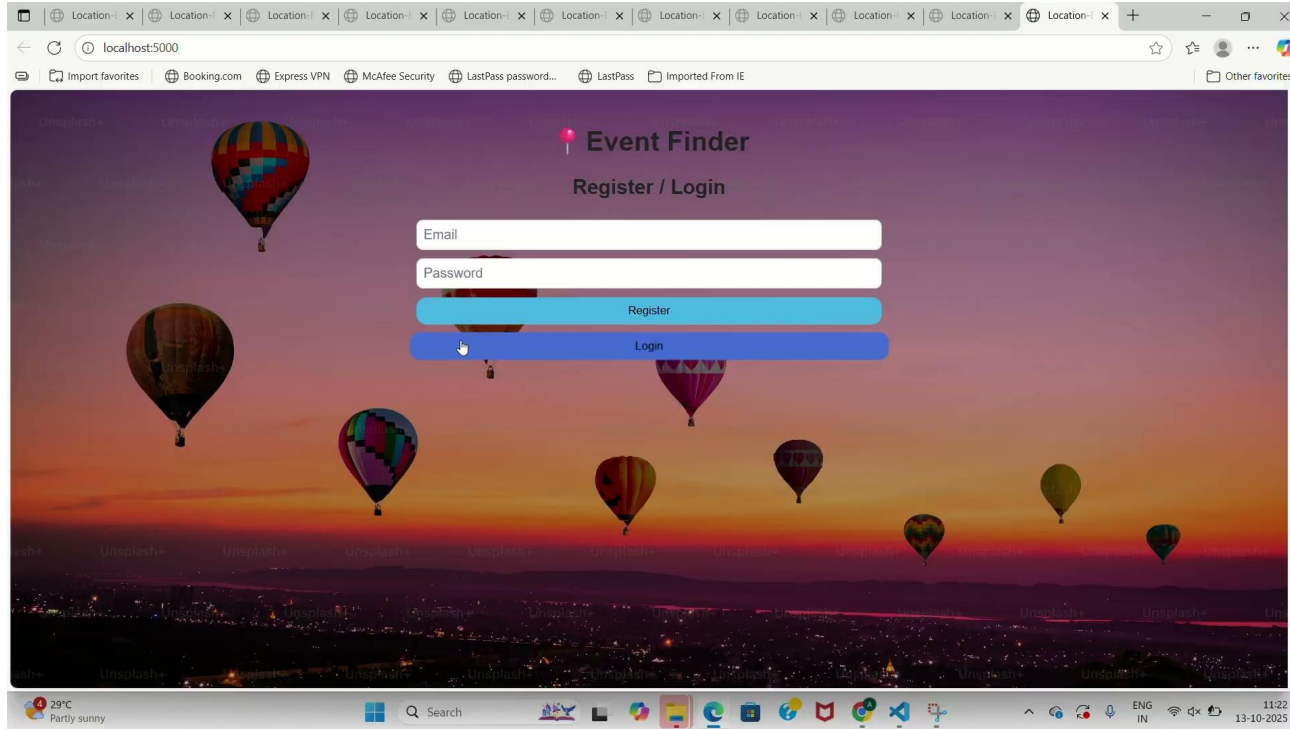
- Real-time displaying of data

## Mitigation

- Implement efficient background update intervals

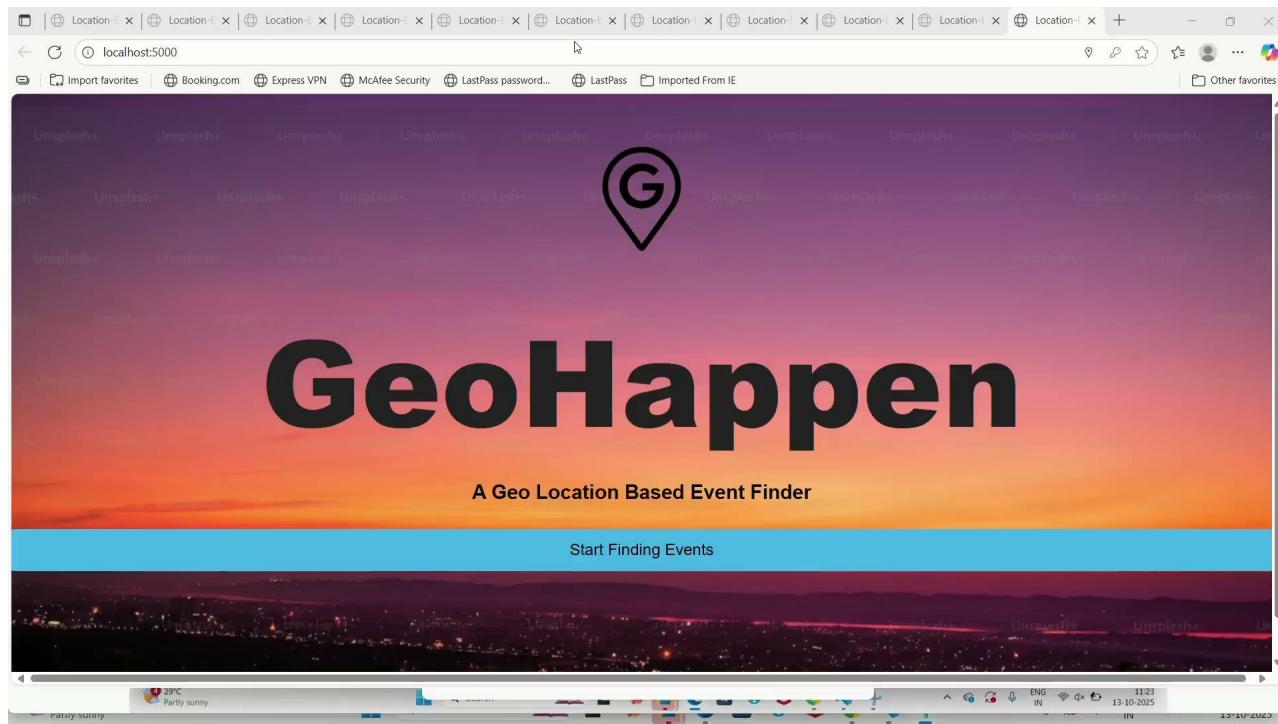


# RESULTS



New user  
registration

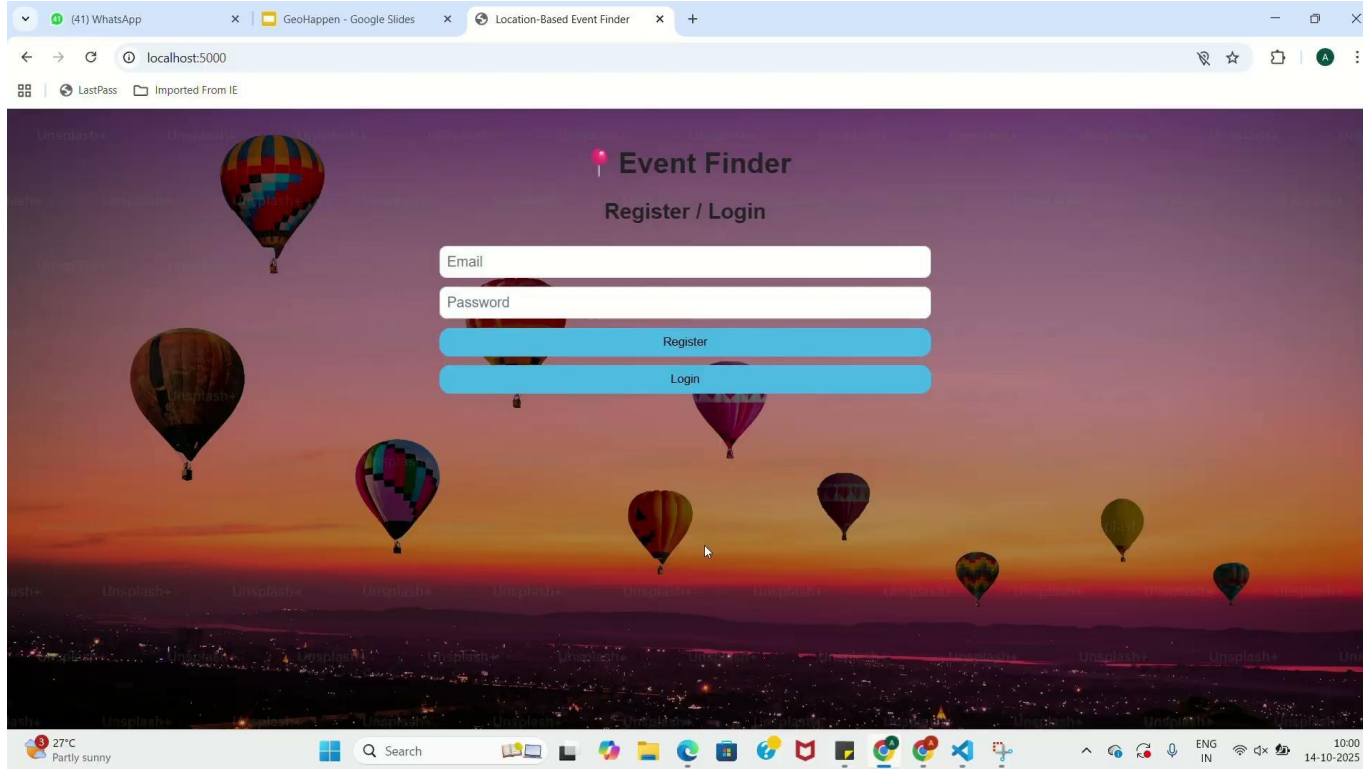
# RESULTS



Adding,  
viewing and  
deleting  
events



# RESULTS



Filtering and recommending events

# CONCLUSION

A geo-location based event finder web application addresses the growing need for a smart, real-time, and location-aware platform that connects users with nearby events based on their interests. By bridging the gap between event organizers and potential attendees, the system enhances local engagement, simplifies event discovery, and promotes community interaction through a seamless digital experience



# REFERENCES

- [1]Sae Majlekar, Chinmayi Deshmane, and Jidnya Kanekar, “Navigating Event Landscapes: A Study of GPS-Based Event Discovery Systems,” International Journal for Research in Applied Science and Engineering Technology, vol. 12, no. 4, pp. 4429–4433, Apr. 2024, doi: <https://doi.org/10.22214/ijraset.2024.61003>.
- [2]P. Afsar, M Faizudheen, M. J. Anikkadan, P. M. Rashad, and U Mohammed Shabeer, “Intelligent Event Finder and Management System,” 2021 Fifth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), pp. 1191–1195, Nov. 2021, doi: <https://doi.org/10.1109/i-smac52330.2021.9640719>.
- 

# REFERENCES

- [3]B. AlBanna, M. Sakr, S. Moussa, and I. Moawad, “Interest Aware Location-Based Recommender System Using Geo-Tagged Social Media,” ISPRS International Journal of Geo-Information, vol. 5, no. 12, p. 245, Dec. 2016, doi: <https://doi.org/10.3390/ijgi5120245>.
- [4]Q. Zhu, S. Wang, B. Cheng, Q. Sun, F. Yang, and R. N. Chang, “Context-Aware Group Recommendation for Point-of-Interests,” IEEE Access, vol. 6, pp. 12129–12144, 2018, doi: <https://doi.org/10.1109/ACCESS.2018.2805701>.
- 